

Assignment 6

Logistic Regression Mini Quiz

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Question 1

A 25-year-old passenger in 2nd class has a predicted probability of survival $P = 0.62$.

Classification: Survived. **Explanation:** Logistic Regression commonly uses a decision threshold of 0.5. Since 0.62 is greater than 0.5, the passenger is classified as **Survived**. Probabilities greater than or equal to the threshold are assigned to class 1 (Survived), while values below the threshold are assigned to class 0 (Did Not Survive).

Question 2

A 45-year-old passenger in 3rd class has $P = 0.38$. **a)** The probability indicates a 38% estimated chance of survival. **b)** If the threshold is changed from 0.5 to 0.4, the prediction still remains **Did Not Survive** because 0.38 is less than 0.4.

Question 3

A 10-year-old passenger in 1st class has $P = 0.85$. **a)** The passenger has an 85% probability of survival and is classified as **Survived**. **b)** Age and passenger class influence the model coefficients. Younger passengers generally have higher survival probability, while passengers travelling in 1st class typically have better survival chances than those in lower classes.

Bonus Challenge

If the coefficient corresponding to **Female** is positive, it means that being female increases the predicted probability of survival compared with the baseline category (male), assuming other variables remain constant.

Conclusion

This assignment demonstrates the interpretation of Logistic Regression probabilities, the importance of decision thresholds, and how model coefficients explain the effect of different features on prediction outcomes.